4/12/18

KABIR KANHA ARORA
1710110165

Department of Computer Science (Shiv Nadar University)
CSD - 205 Discrete Mathematics (End Term Examination)

Date: 4-Dec-2018 **Maximum Marks: 48**

Instructions: **Duration – 2.5 Hours**

1). All questions are compulsory

2). Each question of Part A carries equal marks (2 marks). Each question of Part B carries equal marks (3 Marks).

PART A

1. The number of substrings (Non empty) that can be formed from a character string of length n is

A). n B). n^2 C). $\frac{n(n-1)}{2}$ D). $\frac{n(n+1)}{2}$

2. The number of binary strings of n zeros and k ones such that no two ones are adjacent is

A). ${}^{n+1}C_k$ B). ${}^{n}C_k$ C). ${}^{n}C_{k+1}$ D). None of these

3. Let A be a sequence of 8 distinct integers sorted in ascending order. How many distinct pairs of sequences, B and C are there such that (i) each is sorted in ascending order, (ii) B has 5 and C has 3 elements, and (iii) the result of merging B and C gives A?

A). 2 B). 30 C). 56 D). 256

4. If m identical balls are to be placed in n distinct bags and you are given that $m \geq kn$, where k is a natural number (≥ 1). In how many ways can the balls be placed in the bags if each bag must contain at least k balls?

A). $\binom{m-k}{n-1}$ B). $\binom{m-kn+n-1}{n-1}$ C). $\binom{m-1}{n-k}$ D). $\binom{m-kn+n+k-2}{n-k}$.

Scanned with CamScanner

5. If both 11^2 and 3^3 are factors of the number $a * 4^3 * 6^2 * 13^{11}$, what is the smallest possible value of 'a'?

A) 121 B) 3267 C) 363 D) 33 E) 37

6. Let A be the set of all non-singular matrices over real numbers and let * be the matrix multiplication operator. Then

A). A is closed under * but $< A,* >$ is not a semigroup.

B). $< A,* >$ is a semigroup but not a monoid.

C). $< A,* >$ is a monoid but not a group

D). $< A,* >$ is a group but not an abelian group.

7. Which one of the first order predicate calculus statements given below correctly expresses the following English statement?

Tigers and lions attack if they are hungry or threatened.

A. $\forall x[(tiger(x) \wedge lion(x)) \rightarrow \{(hungry(x) \vee threatened(x)) \rightarrow attacks(x)\}]$

B. $\forall x[(tiger(x) \vee lion(x)) \rightarrow \{(hungry(x) \vee threatened(x)) \wedge attacks(x)\}]$

C. $\forall x[(tiger(x) \vee lion(x)) \rightarrow \{attacks(x) \rightarrow (hungry(x) \vee threatened(x))\}]$

D. $\forall x[(tiger(x) \vee lion(x)) \rightarrow \{(hungry(x) \vee threatened(x)) \rightarrow attacks(x)\}]$

8. Suppose the predicate $F(x,y,t)$ is used to represent the statement that person x can fool person y at time t. which one of the statements below expresses best the meaning of the formula "$\forall x \exists y \exists t(\neg F(x,y,t))$"?

A. Everyone can fool some person at some time.

B. Everyone cannot fool some person all the time.

C. No one can fool everyone all the time.

D. No one can fool some person at some time.

Scanned with CamScanner

9. How many binary relations are there on a set A with n elements? 2^{n^2}

10. In a class of 200 students, 125 students have taken Programming Language course, 85 students have taken Data structure course, 65 students have taken computer organization course, 50 students have taken both Programming language and Data Structures, 35 students have taken both Programming language and Computer Organization; 30 students have taken both Data Structures and Computer Organization; 15 students have taken all the three courses.

How many students have not taken any of the three courses?

(A)15 B). 20 C). 25 D). 35

11. For the set N of natural numbers and a binary operation $f: N \times N \to N$, an element $z \in Z$ is called an identity for f if $f(a,z) = a = f(z,a)$; for all $a \in N$. . Which of the following binary operations have an identity?

(I) $f(x,y) = x + y - 3$ 3

(II) $f(x,y) = \max(x,y)$ 1

(III) $f(x,y) = x^y$

A).I and II only B).II and III only C).I and III only D).None of them

12. The binary relation $R = \{(1,1),(2,1),(2,2),(2,3),(2,4),(3,1),(3,2),(3,3,),(3,4)\}$ on the set $A = \{1,2,3,4\}$ is

A). Reflexive, symmetric and transitive X

B). Neither reflexive, nor irreflexive but transitive

C). Irreflexive, symmetric and transitive X

D). Irreflexive and anti-symmetric X

Scanned with CamScanner

13. Let G be an arbitrary graph with n nodes and k components. If a vertex is removed from G, the number of components in the resultant graph must necessarily lie between

A). k *and* n B). $k+1$ *and* $k+1$ C). $k-1$ *and* $n-1$ D). $k+1$ *and* $n-k$

14. A graph $G=(V,E)$ satisfies $(|E|\leq 3|V|-6)$. The min-degree of G is defined as $\min_{v\in V}\{degree(v)\}$. Therefore, min-degree of G cannot be

A).3 B). 4 C).5 D). 6

15. Let $G_1=(V,E_1)$ and $G_2=(V,E_2)$ be connected graphs on the same vertex set V with more than two vertices. If $G_1\cap G_2=(V,E_1\cap E_2)$ is not a connected graph, then the graph $G_1\cup G_2=(V,E_1\cup E_2)$

A). cannot have a cut vertex

B). must have a cycle

C). must have a cut-edge (bridge)

D). has chromatic number strictly greater than those of G_1 and G_2

PART B

1 Solve the recurrence of Fibonacci Number Series

F (n) = F (n-1) + F (n-2) n>1

F (0) = 0; F (1) =1

2. What is the remainder when $10^{10}+10^{100}+10^{1000}+10^{10000}\ldots+10^{1000000000}$ is divided by 7?

5

3. In how many ways one can wear 5 rings in 4 fingers. Take all possible cases

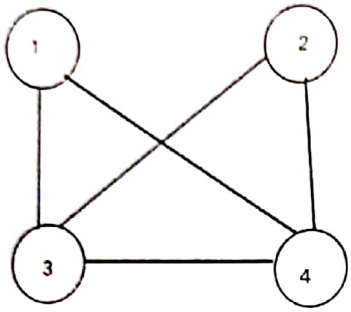
Scanned with CamScanner

No. 5

Cou

4. Five pirates discover a chest containing 100 gold coins. They decide to sit down and devise a distribution strategy. The pirates are ranked based on their experience (Pirate 1 to Pirate 5, where Pirate 5 is the most experienced). The most experienced pirate gets to propose a plan and then all the pirates vote on it. If at least half of the pirates agree on the plan, the gold is split according to the proposal. If not, the most experienced pirate is thrown off the ship and this process continues with the remaining pirates until a proposal is accepted. The first priority of the pirates is to stay alive and second to maximize the gold they get. Pirate 5 devises a plan which he knows will be accepted for sure and will maximize his gold. What is his plan?

5. Find the number of spanning tree for the following graph



6. Here is prerequisites information for some SNU courses:

18.01 → 6.042 ✓

18.01 → 18.03 ✓

8.01 → 8.02 ✓

6.042 → 6.046 ✓

6.001, 6.002 → 6.003 ✓

6.004 → 6.033 ✓

18.01 → 18.02 ✓

6.046 → 6.840 ✓

6.001 → 6.034 ✓

18.03, 8.02 → 6.002 ✓

6.001, 6.002 → 6.004 ✓

6.033 → 6.857 ✓

6.1 Draw a Hasse diagram for the corresponding partially-ordered set

6.2 Suppose that you want to take all the courses. What is the minimum number of terms required to graduate, if you can take as many courses as you want per term?

6.3What is the maximum number of courses that you could possibly take at once?

Scanned with CamScanner